A Web-Based Dashboard for Real-Time Explosibility Analysis of Mine Air

Featuring the Ellicott Explosion Chart with an Intuitive Dark-Theme Interface

Swasti Sundar Nath
Department of Mining Engineering
Indian Institute of Technology (Indian School of Mines) Dhanbad
Dhanbad, Jharkhand, India

Abstract

Underground mining environments are subject to persistent risks of fire and explosion arising from the accumulation of combustible gases such as methane (CH_4), carbon monoxide (CO), and hydrogen (H_2). Accurate, real-time classification of mine air with respect to its explosibility is essential for ensuring worker safety and operational continuity. This paper presents a Python-based web dashboard, developed using the Streamlit framework, that automates the computation of key explosibility parameters — including the Lower Explosive Limit (LEL), Upper Explosive Limit (UEL), excess nitrogen required, and oxygen concentration at the minimum burning limit — and classifies the sampled mine atmosphere using the Ellicott explosion diagram. The application features a professionally designed dark-theme interface with responsive metric cards, colour-coded quadrant visualisation, and an unambiguous zone classification verdict. The dashboard accepts six gas concentration inputs (O_2, N_2, CO_2, CH_4, CO, H_2) and delivers instantaneous graphical and numerical outputs. Validation against a published reference case confirms numerical accuracy. The tool requires no specialist software and is accessible through any web browser, making it immediately deployable in mine safety operations.

Keywords: Mine air; Explosibility; Ellicott diagram; Streamlit; Python; LEL; UEL; Mine safety; Dashboard; Dark UI

1. Introduction

Underground coal and metal mines are characterised by confined atmospheres in which the accumulation of combustible gases — principally methane (CH_4), carbon monoxide (CO), and hydrogen (H_2) — poses a persistent risk of fire and explosion. The fire triangle defines the three necessary conditions for ignition: fuel, oxygen, and a source of initiation. In underground environments, managing the fuel and oxygen components through continuous atmosphere monitoring is the primary means of risk control [1].

The scientific study of explosive gas limits dates to Davy (1816), but the foundational law governing explosive limits of gas mixtures was established by Le Chatelier in 1891 [2]. Subsequent decades saw the development of graphical tools — explosibility diagrams — to provide a visual framework for classifying a sampled atmosphere relative to the explosive zone. The Coward–Jones triangle (1952) [3] was the first widely adopted 2D representation, followed by the simplified Ellicott diagram (1981) [4], the USBM–Zabetakis diagram (1959) [5], the Ternary–Kennedy diagram (1965) [6], and the CCSM–Bardocz diagram (1977) [7].

Despite this body of knowledge, practical application of explosibility analysis in field settings remains largely manual and reliant on offline computation or specialist software. This creates a gap between the theoretical tools available and their day-to-day accessibility to mine safety personnel. Recent work has demonstrated the value of Python-based applications in bridging this gap — for example, Singh et al. [8] developed a Python/Tkinter application for Protective Water Barrier Pillar design, automating regression-based calculations and generating exportable reports for field engineers.

This paper presents a further step in that direction: a fully browser-based mine air explosibility dashboard developed in Python and Streamlit, featuring a purpose-designed dark-theme interface suited to monitoring environments. The application:

- accepts six gas concentration inputs from the user;
- computes LEL, UEL, minimum burning limit, excess nitrogen requirement, and oxygen at the minimum burning limit;
- classifies the atmosphere using the Ellicott explosion diagram with colour-coded quadrant visualisation; and
- presents all results through an accessible, professionally styled dark-theme dashboard with responsive metric cards and unambiguous zone classification.

The paper is structured as follows: **Section 2** reviews the theoretical background; **Section 3** describes the application design and implementation; **Section 4** presents validation results; **Section 5** concludes with future directions.

2. Theoretical Background

2.1 Explosive Limits of Gas Mixtures

The Lower Explosive Limit (LEL) is the minimum concentration of a combustible gas or vapour in air that can sustain explosion in the presence of an ignition source; the Upper Explosive Limit (UEL) is the maximum such concentration. Below the LEL the mixture is too lean; above the UEL it is too rich. Both are equivalent to the Lower and Upper Flammability Limits (LFL, UFL) respectively [2, 3].

For mine air containing multiple combustible gases, the explosive limits of the mixture are calculated using Le Chatelier's mixing law [3]:

$$L_m^{1x} = P_t \div \Sigma\,(P_i / L_i)$$

where P_t is the total fuel content (% vol.), P_i is the concentration of the i-th fuel gas, and L_i is its individual explosive limit. The formula is applied separately for LEL and UEL. Three combustible gases are considered in this application: methane (CH_4, p1), carbon monoxide (CO, p2), and hydrogen (H_2, p3). Gas properties are given in Table 1.

Table 1. Explosive limit and minimum combustion properties of fuel gases [2, 3]

Gas	LEL (%)	UEL (%)	Min. Comb. Limit – Gas (%)	Min. Comb. Limit – O_2 (%)	N^+ to cease combustion (m^3/m^3)
Methane (CH_4)	5.0	14.0	5.9	12.2	6.07
Carbon monoxide (CO)	12.5	74.2	13.8	6.1	4.13
Hydrogen (H_2)	4.0	74.2	4.3	5.1	16.59

2.2 Minimum Burning Limit and Derived Parameters

The minimum burning limit of the mixture (L_{min}) is the minimum fuel concentration below which combustion cannot be sustained. It is computed using the same Le Chatelier formula applied to the minimum combustion limits of each fuel gas:

$$L^{mI}{}_{n} = P_t \div \Sigma\,(P_i / L^{mI}{}_{n,i})$$

The excess nitrogen required (N_{ex}) to render the mixture non-ignitable is:

$$N_e{}^{x} = (L^{mI}{}_{n} / P_t) \times \Sigma\,(N^{+}{}_{i} \cdot P_i)$$

where N_{+i} is the volume of nitrogen required per unit volume of the i-th fuel gas for combustion to cease (Table 1). This figure has direct operational significance: it indicates the quantity of nitrogen that must be injected into a sealed area to suppress explosion risk [2].

The oxygen concentration required at the minimum burning limit is:

$$O_{2,}{}^{mI}{}_{n} = 0.2093 \times (100 - N_e{}^{x} - L^{mI}{}_{n})$$

This threshold defines the oxygen level below which ignition cannot propagate, regardless of fuel concentration [3].

2.3 The Ellicott Explosion Diagram

The Ellicott diagram, introduced by C. W. Ellicott in 1981 as a simplification of the Coward–Jones triangle [4], represents the explosibility state of a gas mixture as a single point in an X–Y coordinate system centred on the nose point of the explosive envelope. The nose point is defined by the coordinates (L_{min}, $O_{2,min}$).

The coordinates of the sample point are:

$$X = P_t - L^{mI}{}_{n}$$
$$Y = O_{2,sa}{}^{m}{}_{ple} - O_{2,}{}^{mI}{}_{n}$$

The four quadrants of the Ellicott diagram correspond to distinct explosibility zones, summarised in Table 2.

Table 2. Ellicott diagram zone classification

Quadrant	Condition	Zone
X > 0, Y > 0	Fuel-rich and oxygen-rich	Explosive Zone
X < 0, Y > 0	Fuel-lean and oxygen-rich	Potentially Explosive — Fuel-lean
X < 0, Y < 0	Fuel-lean and oxygen-deficient	Potentially Explosive — Fuel-rich
X > 0, Y < 0	Fuel-rich and oxygen-deficient	Non-Explosive Zone

The fuel-lean potentially explosive zone indicates that a mixture can become explosive if additional fuel enters the atmosphere; the fuel-rich zone indicates risk if additional air is introduced. Only the first quadrant (X > 0, Y > 0) represents an immediately explosive state [4].

2.4 The Coward–Jones Triangle

The Coward–Jones triangle (1952) [3] is the classical 2D explosibility diagram plotting oxygen concentration on the Y-axis against fuel gas concentration on the X-axis. The explosive triangle is bounded by the LEL and UEL at the base and converges to a nose point defined by the minimum oxygen required for combustion. While the Coward–Jones method offers a more detailed visual representation of the explosive envelope, the Ellicott diagram provides a simpler, operationally practical alternative well-suited to real-time digital interfaces. Integration of the Coward–Jones triangle is identified as a priority for future development (Section 5).

3. Application Development

3.1 Technology Stack and Architecture

The application is developed entirely in Python. The technology stack was selected to prioritise accessibility, rapid deployment, and a high-quality user experience without requiring front-end development expertise:

- Streamlit

The full tech stack is as follows:

- Streamlit: Web application framework that converts the Python script directly into an interactive browser-based interface. No HTML, CSS knowledge is required to deploy it; Streamlit handles server rendering, widget state, and layout automatically.
- NumPy: Numerical computation library used for efficient dot-product operations in the N_e^x calculation and floating-point array handling.
- Matplotlib: Chart generation library used to render the Ellicott explosion chart with colour-coded quadrant fills, axis lines, and the plotted sample point.
- Custom CSS injected via st.markdown(): Overrides Streamlit's default light theme to implement the dark-theme interface, including background colours, card styling, typography, and the zone verdict box.

The application runs as a single self-contained Python script, requiring only

The application runs as a single self-contained Python script, requiring only `streamlit`, `numpy`, and `matplotlib` as dependencies. It is launched with `streamlit run app.py` and is immediately accessible in any web browser at `localhost:8501`.

3.2 User Interface Design

A key contribution of the current version is the deliberately designed dark-theme dashboard interface, which departs from Streamlit's default light appearance. The design rationale is threefold: (1) dark interfaces reduce eye strain in dimly lit monitoring environments common in mine control rooms; (2) colour contrast against a dark background makes the zone classification verdict immediately legible; and (3) the professional aesthetic reinforces the tool's suitability for formal operational use.

The interface is divided into two principal regions:

Left panel (sidebar): **Styled with a dark navy background (**`#111827`), the sidebar presents six labelled numeric input fields for gas concentrations in percent volume: O_2, N_2, CO_2, CH_4, CO, and H_2. The fuel gases (CH_4, CO, H_2) are separated from the inert gases by a horizontal divider, visually grouping the inputs by role. A full-width 'Analyze Sample' button triggers all calculations.

Right panel (main area): **The main content area uses a deep blue-black background (**`#0f172a`) and displays results in a structured vertical layout: a title bar, a brief description of outputs, four metric cards in a responsive column grid, the Ellicott explosion chart alongside two fuel/oxygen metric cards, and finally the zone classification verdict box.

Figure 1 and Figure 2 show the dashboard as rendered in the browser for the reference validation case (Section 4).

3.3 Metric Cards

The four primary computed parameters — LEL, UEL, excess N_2, and required O_2 — are presented as styled HTML metric cards rendered via `st.markdown(unsafe_allow_html=True)`. Each card uses the CSS class `.metric-card` which defines a dark slate background (`#1e293b`), rounded corners, a subtle drop shadow, and centred text layout. The metric value is rendered in a larger font (28px) in sky blue (`#38bdf8`), providing strong contrast against the dark background and immediate visual salience. The relevant CSS is:

```
.metric-card {
    background-color: #1e293b;
    padding: 20px;
    border-radius: 16px;
    text-align: center;
    box-shadow: 0px 4px 12px rgba(0,0,0,0.3);
    margin-bottom: 15px;
}
.metric-value {
    font-size: 28px;
    font-weight: bold;
    color: #38bdf8;
}
```

3.4 Ellicott Chart Implementation

The Ellicott explosion chart is generated using Matplotlib and rendered inline via `st.pyplot(fig)`. The chart occupies the left column of a two-column layout (`st.columns([2, 1])`), with the fuel and oxygen metric cards in the narrower right column, allowing the user to cross-reference the numeric inputs with their chart position at a glance.

The four quadrant regions are filled using `ax.fill_between()` with the following colour scheme, matching the zone classification table:

- Red (alpha 0.3): First quadrant — Explosive Zone
- Green (alpha 0.3): Second quadrant — Potentially Explosive (Fuel-lean)
- Yellow (alpha 0.3): Third quadrant — Potentially Explosive (Fuel-rich)
- Blue (alpha 0.3): Fourth quadrant — Non-Explosive Zone

The sample point is plotted as a large filled circle (`s=180`) with a white edge, providing visibility against all background colours. Grid lines are overlaid at 50% opacity for coordinate reference.

3.5 Zone Classification Verdict

Following the chart, the application renders a full-width zone verdict box using the `.zone-box` CSS class. The background colour of the box is dynamically set to reflect the severity of the zone:

- Red (#dc2626): Explosive Zone — immediate danger
- Green (#16a34a): Potentially Explosive (Fuel-lean) — warning
- Amber (#ca8a04): Potentially Explosive (Fuel-rich) — warning
- Blue (#2563eb): Non-Explosive Zone — safe

The verdict text includes an emoji prefix (□, □, ✅) for rapid visual identification in operational settings where the user may be scanning results quickly.

3.6 Program Flow and Calculation Logic

On receiving valid inputs (sum = 100%), the application executes the following sequence:

1. Compute total fuel content: $P_t = CH_4 + CO + H_2$
2. Apply Le Chatelier's law to compute LEL and UEL of the mixture (Equation 1)
3. Apply Le Chatelier's law to compute L^{mI}_{n} using minimum combustion limits (Equation 2)
4. Compute N_e^{x} using the nitrogen inertion formula (Equation 3)
5. Compute $O_2,^{mI}_{n}$ (Equation 4)
6. Compute Ellicott coordinates: $X = P_t - L^{mI}_{n}$, $Y = O_2 - O_2,^{mI}_{n}$
7. Render metric cards, Ellicott chart, and zone classification verdict

The core calculation block is reproduced below:

```
# Gas property arrays: [CH4, CO, H2]
list1 = [5.0, 12.5, 4.0]      # LEL values (%)
list2 = [14.0, 74.2, 74.2]    # UEL values (%)
list3 = [5.9, 13.8, 4.3]      # Min. combustion limits - gas (%)
list4 = [12.2, 6.1, 5.1]      # Min. combustion limits - O2 (%)
list5 = [6.07, 4.13, 16.59]   # N+ to cease combustion (m3/m3)

pt = ch4 + co + h2             # Total fuel content

# Le Chatelier mixture law
lel_mix = pt / sum(P[i] / list1[i] for i in range(3))
uel_mix = pt / sum(P[i] / list2[i] for i in range(3))
Lmin    = pt / sum(P[i] / list3[i] for i in range(3))

# Inertion and oxygen calculations
Nex    = (Lmin / pt) * np.dot(list5, P)
O2_req = 0.2093 * (100 - Nex - Lmin)

# Ellicott coordinates
x = pt - Lmin        # fuel deviation from nose point
y = O2 - O2_req      # oxygen deviation from nose point
```

Zone classification from Ellicott coordinates:

```
if   x >= 0 and y >= 0:  zone = '□ Explosive Zone'
elif x <  0 and y >= 0:  zone = '□ Potentially Explosive (Fuel-Lean)'
elif x <  0 and y <  0:  zone = '□ Potentially Explosive (Fuel-Rich)'
else:                    zone = '✅Non-Explosive Zone'
```

4. Results and Discussion

4.1 Validation Case

To validate the application, the reference gas composition from Matei et al. [2] was used as input. This composition was originally used to manually demonstrate the Ellicott diagram method, making it a suitable benchmark for numerical verification.

Table 3. Reference input gas composition [2]

Gas	Concentration (% vol.)
O_2	16.00
N_2	74.50
CO_2	2.00
CH_4 (p1)	6.00
CO (p2)	1.00
H_2 (p3)	0.50
Total	100.00

The dashboard computed the following parameters, compared against the reference values:

Table 4. Computed explosibility parameters — dashboard output vs. reference [2]

Parameter	Reference [2]	Dashboard Output	Difference
Total fuel content P_t (%)	7.50	7.500	0.000
LEL of mixture (%)	5.30	5.338	< 0.05
UEL of mixture (%)	17.80	17.712	< 0.10
$L^{mI}{}_{n}$ (%)	6.22	6.220	< 0.01
Excess N_2 required (%)	40.51	40.512	< 0.01
O_2 at minimum burning limit (%)	11.15	11.149	< 0.01

All computed values are in close agreement with the reference, with differences attributable solely to rounding in the published source. This confirms the correctness of the implementation.

4.2 Ellicott Chart Output and Zone Classification

For the validation case, the Ellicott coordinates are:

- X = 7.50 − 6.22 = +1.28 (fuel-rich relative to nose point)
- Y = 16.00 − 11.15 = +4.85 (oxygen-rich relative to nose point)

Both coordinates are positive, placing the sample in the first quadrant — the **Explosive Zone**. This is consistent with the finding of Matei et al. [2], who confirmed through manual Ellicott diagram construction that the same composition falls within the explosive zone. The dashboard renders the sample point in the red quadrant region and displays the verdict: “□ Explosive Zone” in a red zone-box.

4.3 Operational Significance of Outputs

Each output of the dashboard has direct operational relevance for mine safety management:

LEL and UEL (5.30%, 17.80%): Define the explosive window. The relatively wide range reflects the presence of CO and H_2 alongside CH_4, which have higher UELs (74.2%) than methane alone (14.0%).

Excess N_2 required (40.51%): A very substantial nitrogen injection would be needed to render this atmosphere inert — a figure of direct use in planning sealed-area inertisation or ventilation intervention.

O_2 at minimum burning limit (11.15%): Oxygen must be reduced below this threshold to prevent ignition. Normal mine air ($O_2 \approx 21\%$) is far above this value, reinforcing the severity of the classification.

Zone classification verdict: The colour-coded zone box and emoji prefix provide unambiguous, instant visual feedback appropriate for both trained engineers and operational personnel who may not be familiar with the underlying diagram geometry.

5. Conclusion

This paper has presented a web-based mine air explosibility analysis dashboard developed in Python and Streamlit. The application automates the calculation of LEL, UEL, minimum burning limit, excess nitrogen requirement, and oxygen at the minimum burning limit for a three-component fuel mixture (CH_4, CO, H_2), and classifies the sampled atmosphere using the Ellicott explosion diagram. The current version introduces a purpose-designed dark-theme interface with responsive metric cards, a colour-coded quadrant chart, and a dynamically coloured zone verdict box, making the tool suitable for deployment in operational mine monitoring environments.

Validation against the published reference case of Matei et al. [2] confirms numerical accuracy to within rounding error. The tool requires no specialist software, runs in any web browser, and is deployed with a single command.

Future development directions include:

8. Coward–Jones triangle integration: Adding the classical 2D oxygen-vs-fuel diagram alongside the Ellicott chart.
9. USBM–Zabetakis and Ternary–Kennedy diagrams: Extending support to the full suite of standard explosibility diagram methods.
10. Real-time sensor integration: Interfacing with continuous gas monitoring hardware for automatic, sensor-driven classification.
11. Report generation: Automated PDF export of computed parameters and charts, following the approach of Singh et al. [8].
12. Extended gas support: Including ethane (C_2H_6) and propane (C_3H_8) for applicability in oil-bearing coal seam environments.

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